

BADIRA —FMF First-Trimester Preeclampsia Screening

Research notes — What the gold-standard screening measures and what is required to perform it.

1. What Is the FMF Screening Model?

The Fetal Medicine Foundation (FMF) model estimates a woman's risk of developing preeclampsia, especially preterm preeclampsia, at about 11–14 weeks. It combines maternal characteristics and medical/obstetric history with measured biomarkers.

Key point: The result is an individual risk estimate — not a diagnosis of preeclampsia.

2. What Does It Measure?

The main high-performance FMF first-trimester combination is maternal factors + mean arterial pressure (MAP) + uterine artery pulsatility index (UtA-PI) + placental growth factor (PIGF).

PAPP-A can also be included, especially when already available from first-trimester screening, but FMF evidence shows that it does not significantly improve screening when PIGF is already included.

2.1 Maternal Factors

These establish the woman's baseline risk. Examples include age, weight and height, previous pregnancy/preeclampsia history, chronic hypertension, diabetes, method of conception, family history of preeclampsia, and relevant autoimmune disease.

Needed: patient history and/or medical record. No special lab or imaging equipment.

2.2 Mean Arterial Pressure (MAP)

MAP represents average arterial pressure during one cardiac cycle. In FMF screening, blood pressure is measured using a standardized protocol and incorporated into the risk model after appropriate adjustment.

Needed: validated automated blood-pressure monitor, correct cuff size, and standardized measurement technique.

2.3 Uterine Artery Doppler (UtA-PI)

This assesses resistance to blood flow in the uterine arteries supplying the uterus and placenta. The FMF protocol measures the pulsatility index (PI) of the left and right uterine arteries and uses their mean.

Needed: ultrasound machine with color and pulsed-wave Doppler + a trained/certified operator following the FMF protocol.

2.4 Placental Growth Factor (PIGF)

PIGF is a protein involved in placental blood-vessel development. Its level in maternal blood provides information related to placental function and is used as a biochemical marker in the model.

Needed: maternal blood sample + laboratory assay/platform that measures PIGF.

2.5 PAPP-A

PAPP-A (pregnancy-associated plasma protein-A) is a placental protein measured in maternal blood and is often available from first-trimester aneuploidy screening.

Needed: maternal blood sample + compatible laboratory assay.

Important: PAPP-A can be included, but the core high-performance FMF combination is maternal factors + MAP + UtA-PI + PIGF because adding PAPP-A when PIGF is already included does not significantly improve screening performance.

3. How Are the Measurements Combined?

The measurements are not interpreted as separate diagnoses. The model starts with prior risk from maternal factors and history and combines it with the measured biomarkers to calculate a patient-specific risk. Biomarker measurements are standardized as multiples of the median (MoM) with appropriate adjustments before entering the model.

4. Why This Matters for BADIRA

The FMF pathway shows that advanced screening needs more than information entered into an app. Maternal history can be collected digitally, but UtA-PI requires Doppler ultrasound and PIGF/PAPP-A require blood testing and laboratory assays.

6. Requirements at a Glance

Component	Purpose	What is needed
Maternal factors	Baseline risk	History / medical record
MAP	Maternal blood pressure	Validated BP device + correct cuff + standardized protocol
UtA-PI	Uterine artery blood-flow resistance	Doppler ultrasound + trained operator
PIGF	Placental biomarker	Blood sample + laboratory assay
PAPP-A	Additional placental marker	Blood sample + laboratory assay

7. FMF Pathway in One View

1. 11–14 week visit
2. Collect maternal factors/history
3. Measure MAP
4. Measure UtA-PI by Doppler ultrasound
5. Measure PIGF (and PAPP-A if used/available)
6. Combine inputs in FMF competing-risks model
7. Output: patient-specific risk estimate
8. High-risk result → clinical prevention/follow-up; NOT a diagnosis

8. Sources

Fetal Medicine Foundation — Preeclampsia Screening / Certification

Fetal Medicine Foundation — Assessment of Risk for Preeclampsia

O’Gorman et al. — Accuracy of competing risks model in screening for preeclampsia by maternal factors and biomarkers at 11–13 weeks’ gestation (2017)